

Firmware Testing Documentation Template

Document Control

Document Information	Details
Document ID	[Unique ID - e.g., FTD-2025-001]
Related Firmware Update ID	[Reference to Firmware Update Planning document]
Version	[e.g., 1.0]
Created By	[Name and Role]
Creation Date	[YYYY-MM-DD]
Approved By	[Name and Role]
Approval Date	[YYYY-MM-DD]

1. Test Environment Description

1.1 Test Environment Overview

Item	Details
Environment Location	[e.g., Test lab, dedicated rack, etc.]
Isolation Level	[Describe network isolation]
Test Period	[Start date to end date]

1.2 Test Hardware

Hardware Type	Make/Model	Quantity	Firmware Version (Before Test)	Notes
ASIC Miners				
Network Switches				

PDUs				
Environment Sensors				
Other Equipment				

1.3 Test Environment Network

[Describe network configuration, including diagrams if helpful]

1.4 Test Environment Monitoring

[Describe monitoring tools and systems used during testing]

1.5 Data Collection Methods

[Detail how test data will be collected, stored, and analyzed]

2. Test Case Inventory

2.1 Functional Tests

Test ID	Test Name	Description	Priority	Automated/Manual	Owner
FT-001					
FT-002					
[Add more as needed]					

2.2 Performance Tests

Test ID	Test Name	Description	Priority	Automated/Manual	Owner
PT-001					
PT-002					
[Add more as needed]					

2.3 Security Tests

Test ID	Test Name	Description	Priority	Automated/Manual	Owner
ST-001					
ST-002					
[Add more as needed]					

2.4 Compatibility Tests

Test ID	Test Name	Description	Priority	Automated/Manual	Owner
CT-001					
CT-002					
[Add more as needed]					

2.5 Rollback Tests

Test ID	Test Name	Description	Priority	Automated/Manual	Owner
RT-001					
RT-002					
[Add more as needed]					

3. Detailed Test Cases

[For each test case listed in the inventory, provide a detailed section with the following information:]

Test ID: [e.g., FT-001]

Test Name: [Name of the test]

Description: [Detailed description of what the test is checking]

Prerequisites:

- [List any prerequisites for the test]

Test Steps:

1. [Step 1]

2. [Step 2]
3. [Continue as needed]

Expected Results: [Describe the expected outcome in detail]

Actual Results: [To be completed during testing]

Pass/Fail Status:

- ☐ Pass
- ☐ Fail
- ☐ Partially Pass
- ☐ Blocked

Issues Identified: [Document any issues found]

Comments: [Any additional notes or observations]

Tested By: [Name] Date: [YYYY-MM-DD]

4. Performance Baseline and Results

4.1 Baseline Performance Metrics (Before Update)

Metric	Measurement	Method	Date Collected	Notes
Hashrate	[e.g., 95 TH/s avg]			
Power Consumption	[e.g., 3250W avg]			
Efficiency	[e.g., 34.2 J/TH]			
Operating Temperature	[e.g., 65°C avg]			
Rejection Rate	[e.g., 0.5%]			
Fan Speed	[e.g., 60% avg]			
Network Latency	[e.g., 25ms to pool]			
Startup Time	[e.g., 95 seconds]			
[Other metrics as appropriate]				

4.2 Post-Update Performance Results

Metric	Baseline	Post-Update	Change (%)	Within Acceptable Range?	Notes
Hashrate				☐ Yes ☐ No	
Power Consumption				☐ Yes ☐ No	
Efficiency				☐ Yes ☐ No	
Operating Temperature				☐ Yes ☐ No	
Rejection Rate				☐ Yes ☐ No	
Fan Speed				☐ Yes ☐ No	
Network Latency				☐ Yes ☐ No	
Startup Time				☐ Yes ☐ No	
[Other metrics as appropriate]				☐ Yes ☐ No	

4.3 Long-Term Stability Testing

Duration	Start Date/Time	End Date/Time	Issues Observed	Results
24 Hours				
72 Hours				
[Longer if applicable]				

5. Issue Documentation

5.1 Issues Summary

Issue ID	Description	Severity	Test Case	Status	Owner	Resolution
ISS-001						
ISS-002						
[Add more as needed]						

5.2 Detailed Issue Reports

[For each issue, provide detailed information:]

Issue ID: [e.g., ISS-001]

Description: [Detailed description of the issue]

Steps to Reproduce:

1. [Step 1]
2. [Step 2]
3. [Continue as needed]

Expected Behavior: [What should happen]

Actual Behavior: [What actually happens]

Screenshots/Logs: [Reference or include relevant evidence]

Environment: [Specific hardware, configuration where the issue was observed]

Severity:

- ☐ Critical
- ☐ High
- ☐ Medium
- ☐ Low

Impact: [Describe the business impact of this issue]

Recommended Resolution: [Suggestions for fixing the issue]

Resolution Status:

- ☐ Open
- ☐ In Progress
- ☐ Fixed
- ☐ Won't Fix
- ☐ Deferred

6. Test Results Summary

6.1 Test Execution Summary

Test Category	Total Tests	Passed	Failed	Partially Passed	Blocked	Not Run
Functional Tests						
Performance Tests						
Security Tests						
Compatibility Tests						
Rollback Tests						
TOTALS						

6.2 Critical Success Factors

Success Factor	Met?	Evidence/Notes
[e.g., No decrease in hashrate]	☐ Yes ☐ No	
[e.g., Stable operation for 72+ hours]	☐ Yes ☐ No	
[e.g., No increase in rejection rate]	☐ Yes ☐ No	
[e.g., Successful rollback verified]	☐ Yes ☐ No	
[Add more as needed]	☐ Yes ☐ No	

6.3 Open Issues Summary

[Summarize any remaining open issues and their impact]

6.4 Test Results Analysis

[Provide an overall analysis of the test results, including strengths and weaknesses of the firmware update]

7. Recommendations

7.1 Deployment Recommendation

- ☐ Approve for Production Deployment
- ☐ Approve with Conditions (specify below)
- ☐ Reject
- ☐ Defer for Further Testing
- ☐ Conditions/Comments: [Detail any conditions that must be met before deployment or reasons for rejection/deferral]

7.2 Deployment Strategy Recommendations

[Based on testing results, provide recommendations for the production deployment approach]

7.3 Risk Mitigation Recommendations

[Suggest any additional steps to mitigate identified risks]

8. Approval Section

8.1 Test Completion Sign-off

Role	Name	Signature	Date
Test Lead			
Firmware Administrator			
QA Manager			

8.2 Results Review and Approval

Role	Name	Signature	Date
IT/Infrastructure Manager			
Security Officer			
Mining Operations Manager			

8.3 Final Deployment Decision

Role	Name	Decision	Signature	Date
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CTO/IT Director		☐ Approve ☐ Reject ☐ Defer		
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9. Attachments

- ☐ Raw Test Data
- ☐ Performance Graphs
- ☐ Log Files
- ☐ Screenshots
- ☐ Video Recordings
- ☐ Other: _____

Note: This template should be completed for all firmware testing efforts. The depth and extent of testing should be adjusted based on the criticality and scope of the firmware update.